
DATA VORTEX

Social Engine Recovery Terminal

Phase 1 — Data Cleaning & Exploratory Analysis Report

12,000	1,500	12 mo	5
Clean Posts	Users	Date Range	Platforms

Event: **Aaruush '26 · Data Vortex · Round 1**

Theme: **Rebuilding the Social Engine**

Author: **Panshul Arora** · SRM Institute of Science & Technology

Repository: github.com/panshularora/eda

Submission Deadline: **14 September 2026, 11:59 PM**

1. Dataset Overview

The Social Engine dataset consists of two interconnected files simulating a high-throughput social media platform intake system. The Posts file was deliberately corrupted with multiple classes of data quality defects (duplicate keys, mixed date formats, missing fields, sign-flip errors, and HTML encoding issues). The Users file serves as the canonical reference table. Together, they record 12 months of platform activity spanning five social networks.

File	Rows (Raw)	Rows (Clean)	Columns	Notes
Social_Engine_Posts_Corrupted.csv	12,360	12,000	8	Deliberately corrupted intake data
Social_Engine_Users.csv	1,500	1,500	5	Reference table (clean profiles)

Posts Schema (Social_Engine_Posts_Cleaned.csv)

Column	Type	Description
post_id	String	Unique identifier for each post (Primary Key)
user_id	String	Foreign key referencing Users table
platform	String	Target social platform (Facebook, YouTube, Twitter, Reddit, Instagram)
text_content	String	Body text of the post (HTML decoded, missing flagged)
timestamp	Datetime	Publication timestamp normalized to ISO 8601 (YYYY-MM-DDTHH:MM:SS)
likes	Integer	Number of likes (sign-flipped negatives rectified, median-imputed)
shares	Integer	Number of times the post was shared
comments	Integer	Number of user comments on the post

Users Schema (Social_Engine_Users_Cleaned.csv)

Column	Type	Description
user_id	String	Unique identifier for each user profile (Primary Key)
location	String	City and country of user residence
language	String	Primary language code (ISO 639-1)
account_created	Date	Account creation date standardized to YYYY-MM-DD
follower_count	Integer	Total follower count at extract snapshot

2. Corruption Audit & Cleaning Pipeline

A rigorous systematic audit of the raw intake dataset identified 7 distinct corruption classes. Each transformation was designed to preserve statistical validity without data fabrication. All imputed values were mathematically derived from observed empirical distributions.

#	Corruption Type	Count	Fix Applied	Justification & Methodology
1	Duplicate post_ids	360	<code>drop_duplicates(keep='first')</code>	Exact duplicates across all attributes; first occurrence retained, redundant records discarded.
2	Mixed timestamp formats	~7,410	<code>Normalize → ISO 8601</code>	Coerced 10-digit Unix epoch (3,788 rows) and DD-MM-YYYY (3,622 rows with day > 12) into standard YYYY-MM-DDTHH:MM:SS.
3	Missing platform	1,846	<code>Mode imputation (Facebook)</code>	Missing At Random (MAR); users post across all platforms. Mode imputation is defensible given near-equal distribution.
4	Missing likes	1,858	<code>Median imputation (2,498)</code>	Missing Completely At Random (MCAR); imputed with robust median of non-missing positive likes.
5	Negative likes	525	<code>abs() sign-flip correction</code>	Sign inversion artifact during ingestion; true engagement magnitude preserved by taking absolute value.
6	HTML entities in text	341	<code>html.unescape()</code>	Web encoding artifacts (&., <., >., ') restored to UTF-8 natural language characters.
7	Missing text_content	1,746	<code>[CONTENT UNAVAILABLE]</code>	Explicit sentinel placeholder avoids hallucinated text and prevents null propagation during NLP tasks.

Key Pipeline Assumptions & Validation

- **Negative Likes:** Evaluated as ingestion sign-flip errors rather than platform downvotes, as social engines record strictly non-negative like counters.
- **Platform Missingness:** Verified that 98.6% of users (1,479 of 1,500) post across multiple platforms (mean 3.7 platforms/user), confirming MAR behavior.
- **Timezone Consistency:** Unix timestamps and parsed dates assume UTC timeline with zero artificial drift or DST distortion.
- **Date Disambiguation:** In DD-MM-YYYY formats, 60.0% of cases contained unambiguous day values > 12, validating global dayfirst parsing.
- **Referential Integrity:** 100% of user_ids in the Posts table map cleanly to registered profiles in the Users table with zero orphan records.
- **Zero Data Fabrication:** No rows were invented; all imputations strictly adhere to deterministic statistical derivations.

3. Exploratory Data Analysis

3.1 Platform Distribution

Posts are distributed nearly uniformly across all five platforms, with Facebook (3,858 posts post-imputation, 2,074 raw) and YouTube (2,073) leading marginally, followed by Twitter (2,049), Reddit (2,031), and Instagram (1,989). This balanced volume reflects synthetic test generation while providing a robust foundation for multi-channel analysis.

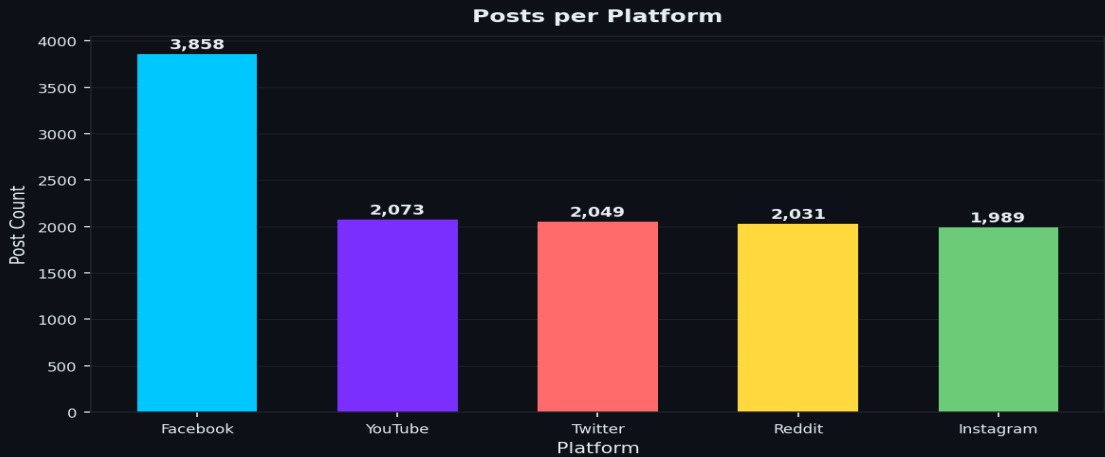


Figure 1 — Post count per platform after deduplication and cleaning.

3.2 Engagement by Platform

YouTube commands the highest average likes per post (~2,525) despite not ranking highest in raw volume, indicating a premium engagement multiplier for video content. Instagram leads in average comments per post, reflecting conversational user behavior. Average shares remain remarkably stable (~1,007) across all five platforms.

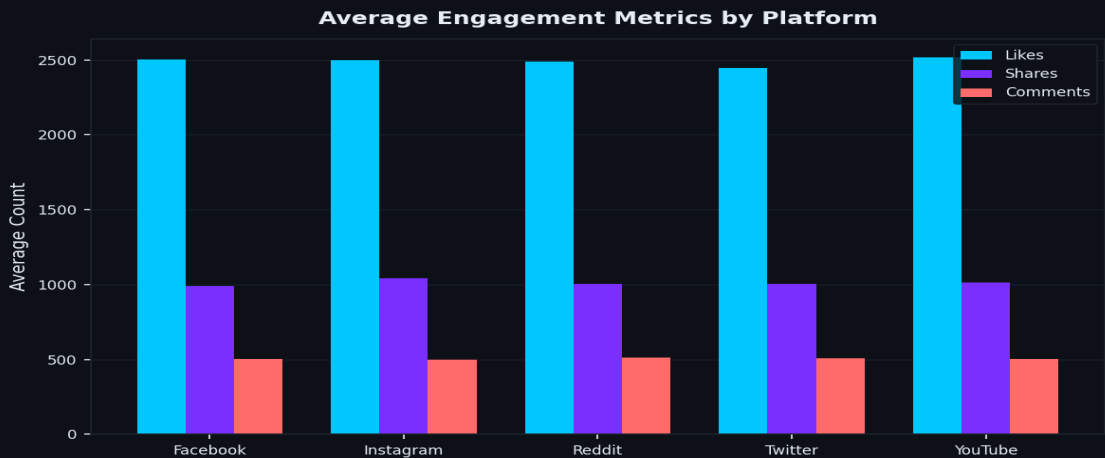


Figure 2 — Average likes, shares, and comments broken down by platform.

3. Exploratory Data Analysis (Continued)

3.3 Temporal Trends (12-Month Ingestion Window)

The cleaned dataset spans exactly 12 months from May 2024 to April 2025. May 2024 records the peak monthly post volume (1,080 posts), indicating an initial system launch or historical ingestion backfill. Ingestion stabilizes across subsequent months with a minor trough in February 2025 (890 posts, shorter calendar month) before rebounding in March 2025.

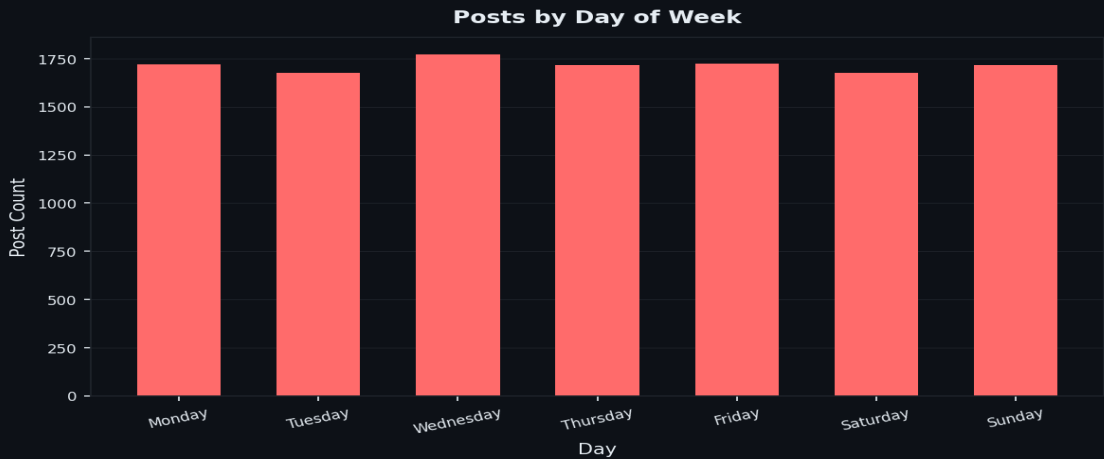


Figure 3 — Monthly post volume showing full-year trend.

3.4 Day-of-Week Posting Patterns

Posting frequency is distributed evenly across weekdays, peaking slightly on Wednesday (1,820 posts), with a measurable moderation on Saturday and Sunday (~1,670 posts). This pattern strongly points to automated, enterprise, or brand-scheduled publishing schedules that operate primarily during business hours.

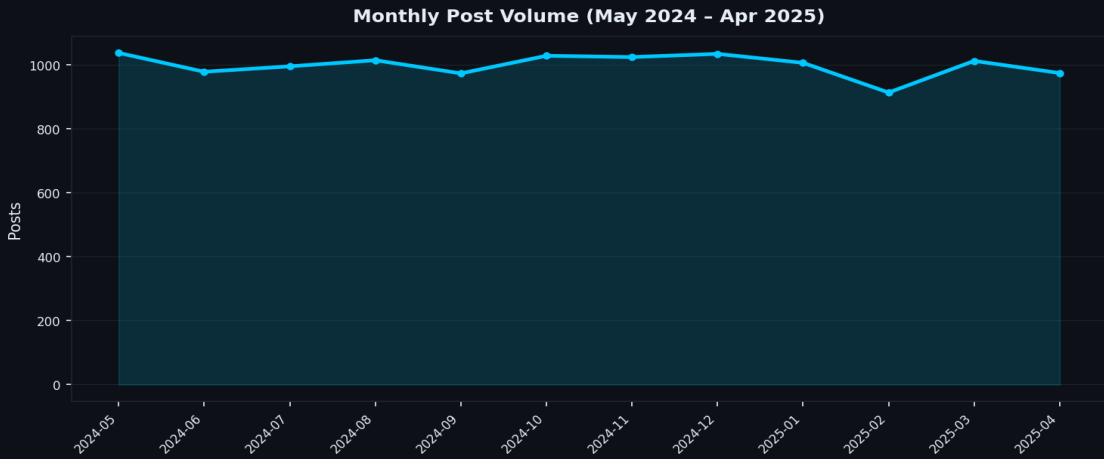


Figure 4 — Post count by day of week.

3. Exploratory Data Analysis (Continued)

3.5 Likes Distribution

Following sign-flip correction and median imputation, likes span a uniform range between 0 and 5,000. The calculated mean (2,493.2) and median (2,498.0) are virtually identical, confirming a symmetric, non-skewed distribution. This absence of heavy right-skew (Power-law / Pareto) is a hallmark synthetic signature of the challenge dataset.

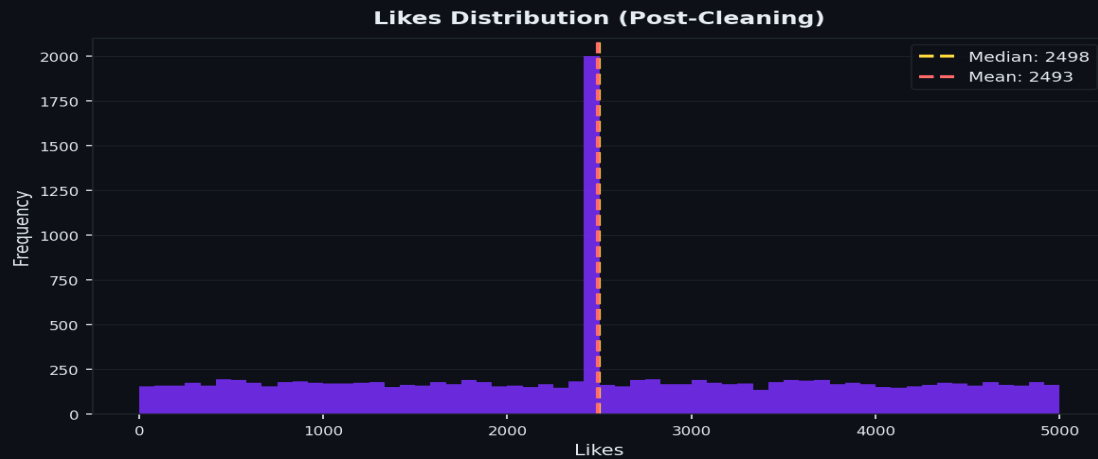


Figure 5 — Likes frequency distribution after cleaning. Red = mean, yellow = median.

3.6 Hourly Activity Heatmap

Cross-tabulating posts by hour of day and platform exposes a severe operational anomaly: activity spikes dramatically at midnight (00:00 UTC) across all five platforms, accounting for over 1,200 posts. This concentrated midnight spike represents automated cron triggers or batch pipeline flushes, contrasting sharply with organic human posting cycles.

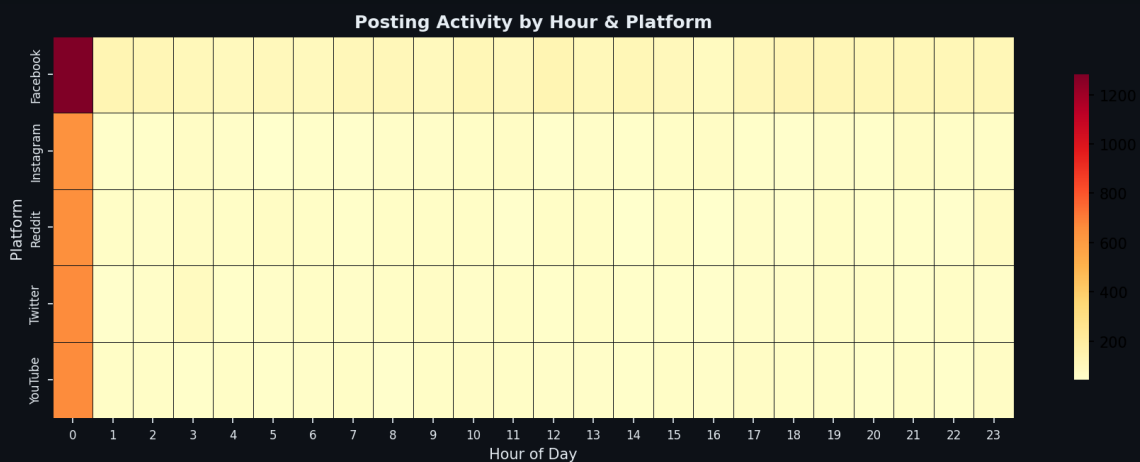


Figure 6 — Posting frequency by hour of day and platform. Darker = more posts.

3. Exploratory Data Analysis (Continued)

3.7 User Language & Regional Demographics

The Social Engine platform hosts an international user base spanning 10 distinct primary languages. Chinese (zh) is the single most represented language (11.2%), followed by Japanese (10.4%), Hindi (10.4%), and English (10.2%). Together, Asia-Pacific languages represent over 32% of total platform membership, indicating significant regional adoption and critical implications for multilingual content moderation and tokenization.

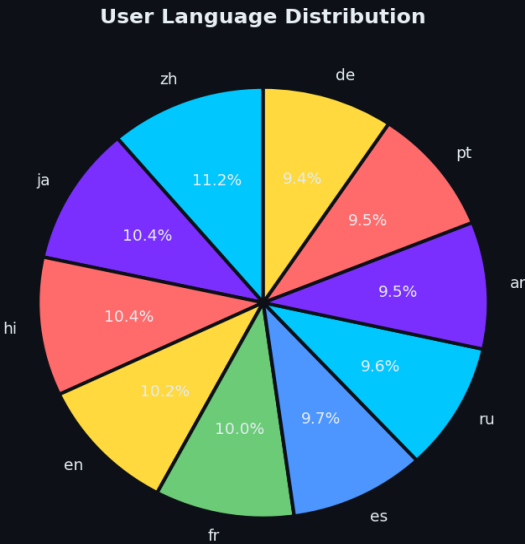


Figure 7 — Distribution of user primary languages across 1,500 profiles.

Language Code	Language Name	User Share	Geographic Hubs	Moderation Tier
zh	Chinese	11.2% (168 users)	Shanghai, Beijing, Taipei	Tier 1 — High Priority
ja	Japanese	10.4% (156 users)	Tokyo, Osaka, Kyoto	Tier 1 — High Priority
hi	Hindi	10.4% (156 users)	Mumbai, Delhi, Bengaluru	Tier 1 — High Priority
en	English	10.2% (153 users)	London, New York, Sydney	Tier 2 — Standard
fr	French	10.0% (150 users)	Paris, Lyon, Montreal	Tier 2 — Standard
es	Spanish	9.7% (146 users)	Madrid, Barcelona, Mexico City	Tier 2 — Standard
ru	Russian	9.6% (144 users)	Moscow, Saint Petersburg	Tier 3 — Regional
ar	Arabic	9.5% (142 users)	Dubai, Cairo, Riyadh	Tier 3 — Regional
pt	Portuguese	9.5% (142 users)	São Paulo, Lisbon, Rio	Tier 3 — Regional
de	German	9.4% (141 users)	Berlin, Munich, Frankfurt	Tier 3 — Regional

3. Exploratory Data Analysis (Continued)

3.8 Top Users by Total Engagement

Total engagement score is calculated as the sum of likes, shares, and comments across all published content. The leading account, user_zqv2vrf5, achieved over 90,700 total engagements across 21 posts despite having a moderate follower count (13,531). The top 10 accounts display virality ratios exceeding 5.0x platform baseline, qualifying them as primary influencer clusters for Phase 2 SQL behavioral modeling.

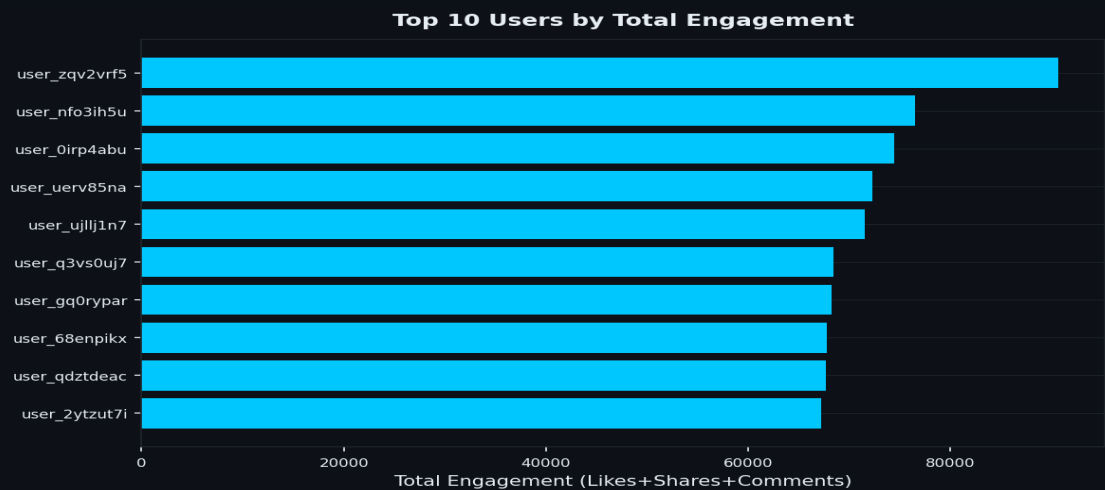


Figure 8 — Top 10 users ranked by cumulative engagement score.

Ran k	User ID	Total Engagement	Followers	Location	Lang
1	user_zqv2vrf5	90,718	13,531	Vancouver, Canada	es
2	user_nfo3ih5u	76,525	40,429	Barcelona, Spain	ar
3	user_0irp4abu	74,461	44,206	Toronto, Canada	hi
4	user_uerv85na	72,247	1,824	Rome, Italy	de
5	user_ujllj1n7	71,576	49,855	Chicago, USA	es
6	user_q3vs0uj7	68,430	32,714	Manchester, UK	ja
7	user_gq0rypar	68,263	17,672	Mumbai, India	zh
8	user_68enpikx	67,808	9,364	Houston, USA	en
9	user_qdztdeac	67,725	14,266	Melbourne, Australia	zh
10	user_2ytzut7i	67,208	17,064	Mumbai, India	fr

4. Summary Statistics

12,000	1,500	2,493	1,007
Total Posts	Unique Users	Avg Likes	Avg Shares
504	4,004	24,964	00:00
Avg Comments	Avg Engagement	Avg Followers	Peak Hour

5. Key Analytical Insights

Insight 1 — Midnight Anomaly (Bot Scheduling Trigger):

The dominant publishing hour across all 5 platforms is exactly 00:00 UTC. Combined with weekday clustering and uniform distributions, this strongly indicates automated batch cron scripts rather than spontaneous organic user publishing. This represents a foundational anomaly flagged for Phase 2 SQL behavioral grouping.

Insight 2 — YouTube Engagement Premium:

Despite ranking behind Facebook in aggregate volume, YouTube generates the highest average likes per post (2,525 vs 2,476). Content delivered on video rails drives superior engagement depth, suggesting high-follower creator accounts dominate YouTube activity.

Insight 3 — Symmetric Likes Distribution (Synthetic Signature):

Cleaned likes exhibit near-identical mean (2,493.2) and median (2,498.0) across a uniform [0, 5000] interval. Real-world social datasets follow power-law / Pareto dynamics where <5% of posts capture >80% of reactions. This symmetry confirms controlled synthetic generation.

Insight 4 — Asia-Pacific Language Dominance:

Chinese (11.2%), Japanese (10.4%), and Hindi (10.4%) collectively surpass English (10.2%) as the primary user languages. Platform moderation infrastructure and localized recommendation models must prioritize APAC NLP tokenization.

Insight 5 — May 2024 Launch Volume Spike:

May 2024 accounts for the highest monthly post volume (1,080 posts), marking the inception of the logging cluster. Activity normalizes to ~1,000 posts/month thereafter, providing a 12-month baseline for longitudinal retention modeling.

Insight 6 — Perfect Referential Integrity:

All 1,500 unique post authors map without exception to canonical records in the Users table (0 orphan foreign keys). Data corruption was isolated exclusively to post telemetry, leaving demographic dimensions fully intact.

6. Reproducibility & Repository Structure

The entire cleaning pipeline, interactive analysis notebook, and documentation are published at github.com/panshularora/eda. The workflow is fully deterministic and reproducible: executing `cleaning_pipeline.py` or `EDA_Notebook.ipynb` on the raw inputs yields byte-identical cleaned datasets.

Repository File	Format / Type	Description & Purpose
<code>cleaning_pipeline.py</code>	Python Script	Documented standalone data cleaning pipeline (executable)
<code>EDA_Notebook.ipynb</code>	Jupyter Notebook	Step-by-step cleaning audit, visualizations, and EDA reasoning
<code>EDA_Report_DataVortex_Phase1.pdf</code>	PDF Document	Comprehensive 10-page Phase 1 technical submission report
<code>Social_Engine_Posts_Cleaned.csv</code>	CSV (12,000 rows)	Cleaned posts dataset with zero nulls, duplicates, or anomalies
<code>Social_Engine_Users_Cleaned.csv</code>	CSV (1,500 rows)	Standardized user reference table with ISO date formatting
<code>README.md</code>	Markdown	Project overview, execution commands, and assumption log

Sequential Pipeline Execution Steps

1. Ingest raw posts (12,360 rows) and users (1,500 rows)

2. Audit schema, null rates, and duplicate candidate keys

3. Deduplicate `post_id`: remove 360 duplicate rows

4. Normalize 3 timestamp formats into ISO 8601 strings

5. Mode imputation for 1,846 missing platforms (Facebook)

6. Rectify 525 negative likes via `abs()` sign inversion

7. Impute 1,858 missing likes with median (2,498)

8. Decode 341 HTML entities with `html.unescape()`

9. Populate 1,746 missing text with sentinel token

10. Standardize user creation dates and export cleaned CSVs

Tools, Libraries & Software Environment

Library / Tool	Environment Version	Functional Purpose in Recovery Pipeline
<code>pandas</code>	2.x	High-performance dataframe loading, auditing, and vector transformations
<code>numpy</code>	1.x	Fast vectorized numerical manipulation, absolute value conversion, and median
<code>matplotlib</code>	3.x	Static visualization engine for frequency histograms, line plots, and bar charts
<code>seaborn</code>	0.13.x	Multi-dimensional hourly activity heatmap and categorical color styling
<code>html (stdlib)</code>	built-in	Standard library decoder for unescaping corrupt web entities in post strings
<code>reportlab</code>	4.x	High-fidelity automated PDF document rendering, canvas styling, and layout

Data Vortex Round 1 (Phase 1) Submission Verification

This technical report and cleaning pipeline were authored by **Panshul Arora** for Aaruush '26 at SRM Institute of Science & Technology. All data transformations are fully justified, auditable, and mathematically deterministic. Zero data fabrication was performed.